

LECTURE 3

The Topology of Metric Spaces: Open and Closed Sets, and Compactness

MATH S4061 | Introduction to Modern Analysis I

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Topology is precisely the mathematical discipline that allows the passage from local to global.

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- 3.1 The algebra of open sets
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Overview. Lecture 2 introduced metric spaces and the open ball (2.3.1). This lecture builds the *topology* they generate: which sets are open—closed under arbitrary unions and *finite* intersections—then the limit (cluster) points that define *closed* sets, the clean duality E open $\iff E^c$ closed, how openness is *relative* to a subspace, and finally *compactness*: the open-cover/finite-subcover property that, unlike open or closed, is *intrinsic* to a set. One method recurs throughout: to show a set is open, write it as a union of balls centred at its own points.

3.1 The algebra of open sets

Definition 3.1.1

Indexed unions and intersections

Let A be an index set and $V_\alpha \subseteq X$ for each $\alpha \in A$. The *union* and *intersection* are

$$\bigcup_{\alpha \in A} V_\alpha = \{x : x \in V_\alpha \text{ for some } \alpha\}, \quad \bigcap_{\alpha \in A} V_\alpha = \{x : x \in V_\alpha \text{ for every } \alpha\};$$

in particular $p \in \bigcap_{\alpha} V_\alpha \iff p \in V_\alpha$ for all α .

Proposition 3.1.2

Unions and finite intersections of open sets

Let (X, d) be a metric space with $V_\alpha \subseteq X$ open for every $\alpha \in A$.

1. $\bigcup_{\alpha \in A} V_\alpha$ is open (any union of open sets is open);
2. $V_{\alpha_1} \cap \cdots \cap V_{\alpha_n}$ is open (any *finite* intersection is open).

Proof 3.1.2

Direct Proof that Unions and Finite Intersections of Open Sets Are Open.

- (1) For the union, let $p \in \bigcup_{\alpha} V_\alpha$, say $p \in V_\beta$. As V_β is open, some $B_{r_p}(p) \subseteq V_\beta \subseteq \bigcup_{\alpha} V_\alpha$; so p is interior and the union is open.
- (2) For a finite intersection, let $p \in V_{\alpha_1} \cap \cdots \cap V_{\alpha_n}$. Each V_{α_i} open gives $r_i > 0$ with $B_{r_i}(p) \subseteq V_{\alpha_i}$. Put $r = \min(r_1, \dots, r_n) > 0$; then $B_r(p) \subseteq V_{\alpha_i}$ for every i , so $B_r(p) \subseteq \bigcap_i V_{\alpha_i}$. The minimum of *finitely* many positive numbers is positive—this is where finiteness is used.

■

[Direct] uses 2.3.3; finiteness enters in (2); cf. R 2.24.

Example 3.1.3

Infinite intersections can fail

The bound $r = \min r_i$ becomes an infimum that may vanish. In $(\mathbb{R}, |\cdot|)$ the open intervals $I_n = (-\frac{1}{n}, \frac{1}{n})$ satisfy

$$\bigcap_{n \geq 1} I_n = \{0\},$$

which is not open in $\mathbb{R}$. So "finite" in 3.1.2(2) cannot be dropped.

Proposition 3.1.4

X and $\emptyset$ are open

In any metric space (X, d) , both X and $\emptyset$ are open in X .

Proof 3.1.4

Direct Proof that X and $\emptyset$ Are Open, the Second Case Vacuously.

- (1) For any $p \in X$ and any $r > 0$, $B_r(p) \subseteq X$, so every point of X is interior: X is open.
- (2) $\emptyset$ has no points, so "every point is interior" holds vacuously: $\emptyset$ is open.

[Direct]

Proposition 3.1.5

Open sets are exactly unions of open balls

A set $V \subseteq X$ is open if and only if it is a union of open balls.

Proof 3.1.5

Constructive Proof that Open Sets Are Exactly Unions of Open Balls.

- (1) If V is open, each $p \in V$ has an $r_p > 0$ with $B_{r_p}(p) \subseteq V$. Every ball lies in V , and each p lies in its own ball, so

$$V = \bigcup_{p \in V} B_{r_p}(p),$$

a union of open balls.

- (2) Conversely any union of open balls is open: each ball is open (2.3.5), and a union of open sets is open (3.1.2).

[Constr] balls are open by 2.3.5; the converse uses 3.1.2.

3.2 Limit points and closed sets

Definition 3.2.1

Cluster (limit) point

Let (X, d) be a metric space and $E \subseteq X$. A point $p \in X$ is a *cluster point*¹ of E if every open ball $B_r(p)$ contains a point of E other than p : for all $r > 0$ there is $q \in E$ with $q \in B_r(p)$ and $q \neq p$. Consequently a finite set has no cluster points.

Definition 3.2.2

Isolated point

A point $p \in E$ is an *isolated point* of E if some ball meets E only at p : there is $r > 0$ with $B_r(p) \cap E = \{p\}$.

¹Also called a *limit point* (Rudin's term) or *accumulation point*. Because every ball holds a point of $E \setminus \{p\}$, it in fact holds infinitely many points of E : a ball with only finitely many would have a least positive distance from p among them, so a smaller ball meets $E \setminus \{p\}$ nowhere (cf. R 2.20).

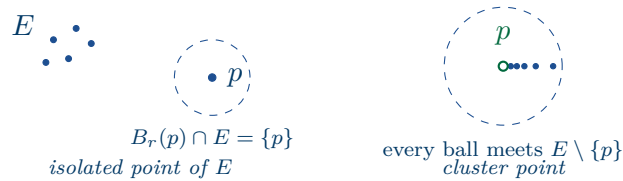


Figure 3.2.3. ISOLATED VERSUS CLUSTER POINTS. Left: p is *isolated* — a ball meets E only at p . Right: p is a *cluster point* — every ball, however small, meets E away from p (3.2.1, 3.2.2).

Definition 3.2.4

Closed set

E is *closed in X* if it contains all of its cluster points: whenever $p \in X$ is a cluster point of E , $p \in E$. To prove E is *not* closed, exhibit one cluster point of E lying outside E .

Example 3.2.5

Cluster points and closedness

In $(\mathbb{R}, |\cdot|)$:

- $E = \{\frac{1}{n} : n \geq 1\}$ has 0 as its only cluster point² (every $B_r(0)$ contains some $\frac{1}{n}$, by the Archimedean property, 1.7.1); since $0 \notin E$, E is *not* closed, whereas $\{0\} \cup \{\frac{1}{n}\}$ is.
- $\mathbb{Z}$ is closed because it has *no* cluster points: each integer is isolated, and any $p \notin \mathbb{Z}$ has $\text{dist}(p, \mathbb{Z}) > 0$, so a small ball about p misses $\mathbb{Z}$. Vacuously, $\mathbb{Z}$ holds all its cluster points.
- Every real is a cluster point of $\mathbb{Q}$ (by density, 1.7.2); $\mathbb{Q}$ is far from closed.

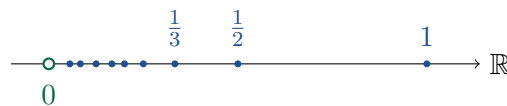


Figure 3.2.6. THE CLUSTER POINT OF $\{1/n\}$. The points $1/n$ bunch arbitrarily close to 0, so every ball about 0 meets $E = \{1/n\}$; yet $0 \notin E$, so E is not closed (3.2.5).

Remark 3.2.7

Open, closed, neither, both

"Closed" is relative: say " E is closed in X ." And open and closed are not opposites—in $(\mathbb{R}, |\cdot|)$ a set may be open, closed, neither, or both (here $p < q$). The interval (p, q) is open; $[p, q]$ is closed; $(p, q]$ is neither; $\mathbb{Q}$ is neither; and $\emptyset$ and X are *both* ("clopen").

Theorem 3.2.8

E open $\iff E^c$ closed

For $E \subseteq X$ with complement $E^c = \{p \in X : p \notin E\}$: E is open in X if and only if E^c is closed in X .

Proof 3.2.8

Proof via Cluster Points that E Is Open $\iff E^c$ Is Closed.

²Each $\frac{1}{n}$ is isolated, and any $p \notin \{0\} \cup E$ has a neighbourhood missing E ; so 0 is the sole cluster point.

- (1) ($\Rightarrow$) Let E be open and $p \in E$. Some $B_r(p) \subseteq E$, so $B_r(p) \cap E^c = \emptyset$ and p is *not* a cluster point of E^c . Thus no point of E is a cluster point of E^c : every cluster point of E^c lies in E^c , i.e. E^c is closed.
- (2) ($\Leftarrow$) Let E^c be closed and $p \in E$. Then $p \notin E^c$; since E^c holds all its cluster points, p is not one, so some $B_r(p) \cap E^c = \emptyset$, i.e. $B_r(p) \subseteq E$. Hence p is interior and E is open. ■

[Direct] cf. R 2.23; recall $(E^c)^c = E$.

Lemma 3.2.9

De Morgan

For any family $\{E_\alpha\}_{\alpha \in A}$ of subsets of X ,

$$\left(\bigcup_{\alpha} E_{\alpha}\right)^c = \bigcap_{\alpha} E_{\alpha}^c, \quad \left(\bigcap_{\alpha} E_{\alpha}\right)^c = \bigcup_{\alpha} E_{\alpha}^c.$$

Proof 3.2.9

Proof of De Morgan's Laws by Element-Chasing.

- (1) For the first identity,
- $$p \in \left(\bigcup_{\alpha} E_{\alpha}\right)^c \iff p \notin \bigcup_{\alpha} E_{\alpha} \iff p \notin E_{\alpha} \forall \alpha \iff p \in E_{\alpha}^c \forall \alpha \iff p \in \bigcap_{\alpha} E_{\alpha}^c.$$
- (2) The second follows by replacing each E_{α} with E_{α}^c and taking complements. ■

[Direct] cf. R 2.22.

Proposition 3.2.10

Intersections and finite unions of closed sets

Let $F_{\alpha} \subseteq X$ be closed in X for every $\alpha \in A$.

1. $\bigcap_{\alpha \in A} F_{\alpha}$ is closed (any intersection of closed sets is closed);
2. $F_{\alpha_1} \cup \dots \cup F_{\alpha_n}$ is closed (any *finite* union is closed).

Proof 3.2.10

Proof by Dualising the Open Case that Intersections and Finite Unions of Closed Sets Are Closed.

- (1) For the intersection, de Morgan gives $(\bigcap_{\alpha} F_{\alpha})^c = \bigcup_{\alpha} F_{\alpha}^c$. Each F_{α}^c is open (3.2.8), and a union of open sets is open (3.1.2); so the complement is open and $\bigcap_{\alpha} F_{\alpha}$ is closed (3.2.8).
- (2) For a finite union, $(F_{\alpha_1} \cup \dots \cup F_{\alpha_n})^c = F_{\alpha_1}^c \cap \dots \cap F_{\alpha_n}^c$ is a *finite* intersection of open sets, hence open (3.1.2); so the union is closed. (Infinite unions can fail, dually to

3.1.3.)

[Direct] via 3.2.8, 3.2.9, 3.1.2; cf. R 2.24. A proof straight from the cluster-point definition (the board's exercise) also works. ■

3.3 The subspace topology

Proposition 3.3.1

Relative openness

Let (X, d) be a metric space, $Y \subseteq X$ a subspace, and $E \subseteq Y$. Then E is open in Y if and only if $E = Y \cap G$ for some G open in X . (Here the ball in Y is $B_r^Y(p) = \{x \in Y : d(x, p) < r\} = B_r(p) \cap Y$.)

Proof 3.3.1

Proof of Relative Openness by Carving Y -balls from X -balls.

- (1) ($\Leftarrow$) Suppose $E = Y \cap G$, G open in X . For $p \in E$ we have $p \in G$, so some $B_{r_p}(p) \subseteq G$; intersecting with Y , $B_{r_p}^Y(p) = B_{r_p}(p) \cap Y \subseteq G \cap Y = E$. Thus p is interior in Y , and E is open in Y .
- (2) ($\Rightarrow$) Suppose E is open in Y : each $p \in E$ has $r_p > 0$ with $B_{r_p}^Y(p) \subseteq E$. Put $G = \bigcup_{p \in E} B_{r_p}(p)$, open in X as a union of X -balls (3.1.5). Then

$$G \cap Y = \bigcup_{p \in E} (B_{r_p}(p) \cap Y) = \bigcup_{p \in E} B_{r_p}^Y(p) = E.$$

[Direct] cf. R 2.30. ■

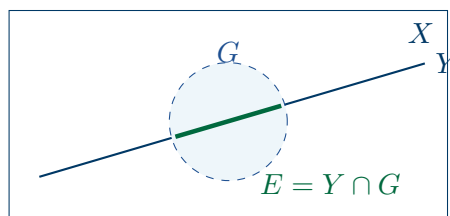


Figure 3.3.2. OPEN IN A SUBSPACE. E is open in Y exactly when $E = Y \cap G$ for some G open in the whole space X : an X -ball G meets the subspace Y in the relative set E (3.3.1).

Remark 3.3.3

Openness by balls at its own points

The recurring method, used in 3.1.5 and 3.3.1: to prove a set is open, write it as a union of open balls centred at its own points.

3.4 Compactness

Definition 3.4.1

Open cover; compact set

Let (X, d) be a metric space and $K \subseteq X$. An *open cover* of K is a family $\{V_\alpha\}$ of sets open in X with $K \subseteq \bigcup_\alpha V_\alpha$. The set K is *compact* if every open cover of K admits a *finite subcover*: indices $\alpha_1, \dots, \alpha_n$ with $K \subseteq V_{\alpha_1} \cup \dots \cup V_{\alpha_n}$.³

Example 3.4.2

Two non-compact sets

In $(\mathbb{R}, |\cdot|)$:

- $\mathbb{R}$ is not compact: the cover $V_n = (-n, n)$ ($n \geq 1$) has $\bigcup_n V_n = \mathbb{R}$, but any finite subfamily lies in V_N with $N = \max n_i$, missing all $|x| \geq N$.
- $(-1, 1)$ is not compact: $V_n = (-1 + \frac{1}{n}, 1 - \frac{1}{n})$ ($n \geq 2$) cover $(-1, 1)$, yet any finite subfamily is contained in the largest V_N , missing points near ± 1 .



Figure 3.4.3. AN OPEN COVER WITH NO FINITE SUBCOVER. The intervals $V_n = (-1 + \frac{1}{n}, 1 - \frac{1}{n})$ cover $(-1, 1)$, but each stops short of ± 1 ; a finite subfamily reaches only the largest V_N — so $(-1, 1)$ is not compact (3.4.2).

Proposition 3.4.4

Compactness is intrinsic

Let $K \subseteq Y \subseteq X$, where (X, d) is a metric space and Y carries the subspace metric $d|_{Y \times Y}$. Then K is compact in Y if and only if K is compact in X . So one may say simply " K is compact," with no ambient space.

Proof 3.4.4

Proof that Compactness Is Intrinsic by Trading Y -covers for X -covers.

- | | |
|--|--|
| <div style="border-left: 1px solid black; height: 100px; margin-left: 5px;"></div> | <p>(1) ($\Leftarrow$) Assume K compact in X. Let $\{V_\alpha\}$ be a Y-open cover of K. By 3.3.1 each $V_\alpha = G_\alpha \cap Y$ with G_α open in X, and $\{G_\alpha\}$ covers K. By X-compactness, $K \subseteq G_{\alpha_1} \cup \dots \cup G_{\alpha_n}$; intersecting with Y (and using $K \subseteq Y$), $K \subseteq V_{\alpha_1} \cup \dots \cup V_{\alpha_n}$.</p> <p>(2) ($\Rightarrow$) Assume K compact in Y. Let $\{G_\alpha\}$ be an X-open cover of K. Set $V_\alpha = G_\alpha \cap Y$, open in Y (3.3.1) and covering K. By Y-compactness, $K \subseteq V_{\alpha_1} \cup \dots \cup V_{\alpha_n} \subseteq$</p> |
|--|--|

³The naive candidate "closed and bounded" depends on the ambient space and the metric; the open-cover definition is *intrinsic* (3.4.4) and survives passage to continuous images, which is why it is the right notion.

$$G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}.$$

[Direct] uses 3.3.1; cf. R 2.33.



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